Better Pooled Features, Worse Transfer: Training Divergence in Graph In-Context Learning

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Abstract

A graph in-context learner can transfer poorly because its representations lack useful information or because subsequent computation fails to use that information. These explanations imply different remedies, but a single end-to-end score cannot distinguish them. We study their divergence during PRODIGY pretraining using saved checkpoints, fixed target episodes and support-fitted readouts at distinct computational stages. Across nine source models trained with the corrected production sampler, advancing from 100 to 2500 updates improves a fixed ridge readout of pre-projection pooled features on TwiBot20 while degrading full-model ranking. The opposite signs hold within episodes for every source in both evaluation streams: source-mean changes are +4.21/+4.36 versus -9.00/-10.14 AUC points. The same training improves full-model ranking on political, election and page classification for every source. Training interventions further separate intermediate improvement from successful end-to-end repair: freezing the initial center/pool projections improves the Facebook U1 probe consistently but not full inference, while gradient isolation fails its declared repair criterion. Matched support/query interventions identify a class-reference pathway and show why the sign of a context ablation cannot be interpreted as encoder quality alone. Our contribution is a target-dependent training phenomenon and its computationally resolved diagnosis, not a universally harmful solver, a successful adaptation method or a complete predictive mechanism.

1. Introduction

More effective pretraining need not produce a better transferred classifier. This is not merely the familiar observation that some source datasets transfer better than others. On fixed target examples, additional training may make an intermediate representation more useful to a fixed classifier while making the model's own predictions worse. The distinction matters for interpreting source quality, training budgets and proposed repairs: an end-to-end decline does not by itself establish that useful graph information has been lost everywhere.

We investigate this possibility in graph in-context learning. PRODIGY processes sampled neighborhoods, transforms and aggregates their representations, and communicates over a task graph connecting examples to class references. Each stage can change which information a downstream readout can exploit. We keep the target episodes fixed across checkpoints and evaluate the same support-only ridge rule at different stages alongside full inference.

The clearest result is a target-dependent divergence. On a derived TwiBot20 retweet graph, all nine corrected-sampler source models improve the ridge readout of pre-projection pooled features between updates 100 and 2500, while all nine full models decline. This is not solely an effect of pooling scores from different episodes: it holds for mean within-episode AUC. Conversely, all nine full models improve on political, election and page classification. Neither source-wide failure nor universally harmful training describes these observations.

Simple classifiers on pretrained features and the separation of training from adaptation are established ideas in few-shot learning (Tian et al., 2020; Luo et al., 2023). We do not claim those ideas as new. Our empirical contribution is the source-broad reversal between an intermediate readout and full transfer over a matched training trajectory, its target specificity, and evidence that improving an intermediate probe through training is not sufficient to repair the full model. Role-resolved experiments provide a complementary diagnostic: context can affect the class reference without changing the query, and its measured utility depends on the inference rule. These findings constrain explanations of graph transfer without turning unsuccessful repairs into methods.

2. Separate stages, preserve comparisons

An episode contains labeled supports and queries, each expanded into a sampled subgraph. In the evaluated implementation, background graph processing produces node features. We observe a pooled representation, S0_pool, immediately before the learned reset_mlp_m projection. U1_pre_meta is the actual example vector after the subsequent projection/aggregation computation. A metagraph then updates example and class representations, and full inference uses scaled query-to-class cosine scores.

These stage names are substantive. Improvement at S0_pool does not establish improvement of the entire encoder: U1 may move in the opposite direction. Likewise, substituting a probe changes more than one native operation and is not a causal ablation of the metagraph alone.

The fixed ridge readout row-normalizes support and query vectors, fits one-hot support targets with regularization lambda=1 and no intercept, and scores the queries. It never fits query labels or selects hyperparameters on target outcomes. Prototype comparisons instead normalize individual vectors, average supports within class, normalize class means and use cosine similarity.

The corrected-sampler trajectory includes nine source runs with seed 0 and checkpoints 100, 300, 900 and 2500. Our reported endpoint contrast uses 100 and 2500; it does not assert monotonic changes at intervening checkpoints. Training sidecars verify completed updates and architecture/configuration, and cached target tensors are matched across checkpoints. The corrected recipe changes retention and role assignment jointly relative to historical training; old/new recipe contrasts are not exact matched-training causal comparisons.

Each of five targets has two streams of 128 binary episodes, with ten supports per class. Query occurrences per stream are 3,072 for political and TwiBot20, 1024 for pages, and256 for election and suspension. Accounts recur; streams are not additional training seeds. We report both globally pooled AUC under the declared semantic orientation and mean within-episode AUC where specified. For pages, classes are episode-local category pairs. Source consistency is not a confidence interval or evidence of broad target diversity.

The targets are derived social-media graphs rather than interchangeable standard benchmarks. Political labels include heuristics; suspension is not bot status; TwiBot20 uses a reconstructed retweet graph, not its original follow graph. The public knowledge-graph experiment below supplies a boundary under another setting, not a replication of the social mechanism.

3. Training divergence is systematic across sources, specific to targets

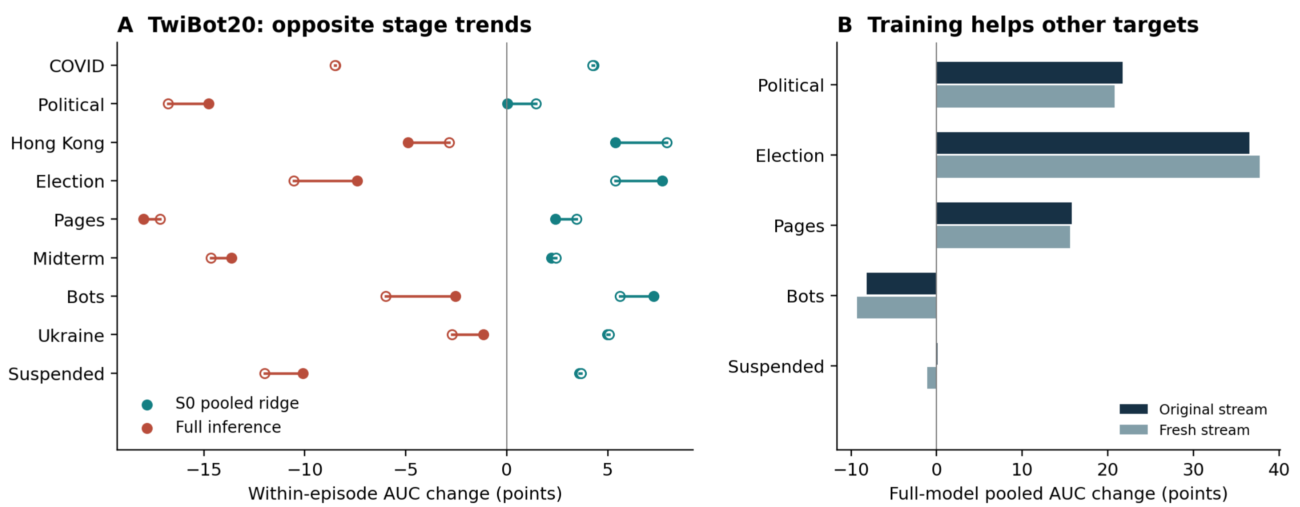

Figure 1: Training from update 100 to 2500 has target- and stage-dependent effects. A: source-specific mean within-episode AUC changes on TwiBot20; filled/open points are original/fresh streams, not error bars. B: full-model pooled AUC changes averaged over nine sources on every target. The two panels deliberately report different AUC estimands, labeled on their axes. Each stream contains 128 episodes per target; nine corrected-sampler source runs use one training seed.

Figure 1 shows the nine-source endpoint changes, retaining both streams and labeling the exact representation stage. On TwiBot20, S0_pool ridge improves while full-model AUC declines for all nine sources on both streams. Mean within-episode AUC changes are:

Readout	Original stream	Fresh stream
S0_pool ridge	+4.21 points	+4.36 points
Full inference	-9.00 points	-10.14 points

This rules out a purely cross-episode scoring explanation for the opposite signs. It does not establish significance for each source: the smallest pooled probe gain is 0.0434 points. U1 ridge declines in eight of nine original-stream cells and seven of nine fresh-stream cells. The divergence therefore spans processing before and after U1; it cannot be assigned wholly to the metagraph. U1 can nevertheless remain better than S0_pool in absolute performance while declining over training. Stage scores are not an additive decomposition.

The strongest counterexample is the same training evaluated on other targets. Across the nine sources, full-model pooled AUC changes average:

Target	Original stream	Fresh stream	Source direction
Political	+21.70	+20.82	All nine improve in both streams
Election	+36.53	+37.79	All nine improve in both streams
Pages	+15.80	+15.62	All nine improve in both streams
TwiBot20	-8.18	-9.30	All nine decline in both streams
Suspension	+0.07	-1.13	Mixed

Values are AUC points. This pattern identifies a conflict in transfer across targets, not an intrinsically harmful architecture or training procedure. Why these computations preserve one target's useful distinctions while weakening another's remains an explanatory question.

4. Better intermediate readouts do not guarantee repair

An 18-model experiment compares free training with the initial center/pool projections frozen, crossing three sources and three initialization seeds. Specifically, it fixes the weights and biases of `reset_mlp_c` and `reset_mlp_m`; the background convolution and metagraph solver still train. Each pair shares exact initialization and all consumed training inputs; frozen tensors remain fixed throughout training while other components adapt. The declared primary was consistent Facebook full-model improvement. It fails. Nevertheless, the Facebook U1 ridge probe improves in all nine matched pairs on both streams. That secondary result shows the discrepancy can persist when downstream computation has adapted during training, not only after an abrupt parameter swap. It does not convert the failed full-model criterion into a successful repair. The earlier S0_pool ridge probe instead decreases in 17 of 18 source-seed-stream comparisons. Thus the result depends on which representation stage is evaluated; it is not a uniform improvement of the encoder. U1 is the same named stage as above, but the sources, sampler protocol and target differ from the corrected-sampler TwiBot trajectory and must not be pooled as one causal experiment.

A separate eight-arm experiment compares native, joint native-plus-ridge, gradient-isolated and ridge-only training under blocked and interleaved four-source schedules. Initial weights and per-source examples match. Isolated and ridge-only encoder trajectories diverge numerically, so they are not identical learned encoders. On the fixed four-target supported panel, isolated full inference loses in mean AUC to native, joint and its own U1 ridge under both schedules. The proposed method fails its criterion and is not expanded.

Native/joint training nevertheless yields stronger U1 ridge performance than ridge-only in aggregate. Thus a learned solver may benefit the training system without being the best inference procedure. This is supporting evidence rather than a new general principle; it is one-seed evidence with target exceptions. For blocked political classification, native full inference beats its ridge readout, ruling out a uniformly unnecessary solver.

5. Context changes references as well as queries

The training divergence does not tell us how any individual graph-context intervention will behave. In historical Hong Kong-trained models, removing background edges only from support subgraphs can improve political/page performance while leaving query tensors fixed. The operation retains sampled nodes, features and pooling connections. Swapping the resulting class vectors reproduces the changed scores, locating a support-to-reference pathway under the evaluated one-metagraph-layer, frozen-normalization protocol. This is computational localization, not an explanation of its utility.

Matched page trajectories make the interpretive limit concrete. Across three seeds and two streams, suppression changes native mean episode AUC by -3.41 points at update 100 and +0.89 at 2500. The fixed prototype instead benefits at both checkpoints: +7.93 and +1.76 points, positive in all six cells at each step. Both native conditions improve absolutely. The native sign reversal cannot be equated with useful context information becoming harmful; a fixed readout already exploits the suppressed representations early. The nominated prediction that context would remain helpful to the prototype was contradicted.

The broader intervention map includes other affected sources and reversals across targets. It is a scope map, not a sign predictor. Public evidence reinforces the boundary: an original-style Wiki-to-FB15K-237 model achieves 73.8% accuracy, falling to 39.9% or 39.0% with support or query suppression. Both roles can be useful. Public training-mode normalization means queries are not fixed under those interventions; social fixed-query mediation does not generalize by assertion. The public replay required an explicit accuracy-equivalence amendment after its original logit tolerance failed.

6. Implications and remaining explanation

The relevant comparison is not simply which readout wins at one checkpoint. It is whether learning improves accessible intermediate distinctions and end-to-end transfer together. Here those trajectories can separate systematically across sources while agreeing on other targets. Reporting only final scores would hide that distinction; reporting only favorable probes would hide the failure of the complete model and the unsuccessful repairs.

The evidence supports a computationally resolved phenomenon study of this implementation and target panel. It does not establish an information-theoretic loss, a universal graph-ICL failure, a deployable intervention selector, or a property predicting unseen target behavior. Generic representation/readout comparisons are prior art. The outstanding contribution-strengthening question is a graph-specific account of the target dependence, rather than another unselected control or a stronger claim about the failed isolation method.

References to integrate

- Huang et al., 2023. PRODIGY: Enabling In-context Learning Over Graphs.
- Tian et al., 2020. Rethinking Few-Shot Image Classification: a Good Embedding Is All You Need?
- Luo et al., 2023. A Closer Look at Few-shot Classification Again.
- Hou and Sato, 2022. A Closer Look at Prototype Classifier for Few-shot Image Classification.

The existing dataset citations, ethics, AI-use and restricted-artifact statements must be retained and reconciled during integration. Reviewer access remains TBD. No submission, release permission or human-author approval is asserted here.